Policy Gradient Theorem

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Trajectory is like a complete storyline of an episode, holding the full history of what the agent saw(state), did(action), and received(reward). Trajectories are commonly represented by τ and is represented as a list of steps, where each step is a tuple of (state, action, reward) pair -

$$\tau = [(s_0, a_0, r_0), (s_1, a_1, r_1).....(s_t, a_t, r_t)] \tag{1}$$

Here, at time step t the agent observes state s_t and takes action a_t. This action results in reward r_t from the environment. If reward is delayed and received at the end of the episode all r_t's before the last step can be 0 or undefined.
Our objective function is to maximize the expected reward over all the trajectories in the environment. Here R is the return function which provides the reward for each trajectory.

$$J(\pi_\theta) = \mathbb{E}_\tau[R(\tau)] \tag{2}$$

Now, lets expand the expectation in Eq. 2. An expectation of a distribution is the long-term average value of a random variable. It is calculated by multiplying each possible value by its probability and then summing the results. We will do the same here i.e. for all the trajectories that are possible in the environment under a given policy we multiply the probably of the trajectory(τ) with the returns generated over that trajectory τ

$$J(\pi_\theta) = \int_\tau \mathbb{P}(\tau|\theta)R(\tau) \tag{3}$$

Since our goal is to maximize the expected return we have to update the policy in the direction that maximizes it i.e. perform gradient ascent. To achieve this we find the gradient of the objective function with respect to the policy parameters θ. The idea is to take steps proportional to the positive gradient so as to move closer to a local maximum.

$$\Delta_\theta J(\pi_\theta) = \int_\tau \Delta_\theta \mathbb{P}(\tau|\theta)R(\tau) \tag{4}$$

One thing to notice here is that solving the above equation means computing gradients and rewards over all possible trajectories — which quickly becomes infeasible in large or continuous environments. A more practical way around this is to express it as an expectation, allowing us to approximate it using Monte Carlo sampling, rather than exhaustively computing everything. This is where the log-derivative trick Eq.5 is used. It is used to rewrite the gradient of an expectation in a form suitable for Monte Carlo approximation.

$$\Delta \log(f(x)) = \frac{\Delta f(x)}{f(x)} \tag{5}$$

Essentially we want to replace the gradient of probability distribution function in Eq.4 with the following

$$\Delta \mathbb{P}(\tau|\theta) = \mathbb{P}(\tau|\theta)\Delta \log \mathbb{P}(\tau|\theta) \tag{6}$$

Adding this back into the main objective will give us the following -

$$\Delta_\theta J(\pi_\theta) = \int_\tau \mathbb{P}(\tau|\theta)\Delta_\theta \log \mathbb{P}(\tau|\theta)R(\tau) \tag{7}$$

If you look closely, this equation represents a weighted sum of some function values, where each term is scaled by its probability. This is exactly how an expectation is defined — the average outcome when sampling from a distribution. And we all know how to solve an expectation, by estimating it using samples(Monte Carlo estimate). So, essentially instead of summing over all possible trajectories (which is impossible), we can sample a few trajectories from the policy and take their average reward-weighted gradient to get the expectation, giving us the following equation –

$$\Delta_\theta J(\pi_\theta) = \mathbb{E}_{\tau \sim \pi_\theta}[\Delta_\theta \log \mathbb{P}(\tau|\theta) R(\tau)] \tag{8}$$

But this doesn't end here, we can simplify the equation even more. Lets focus on the term $\Delta_\theta \log \mathbb{P}(\tau|\theta)$ which is the probability of the trajectory under policy π_θ. This probability is computed as the product of probability of each step which is, the probability of taking action a_t in s_t and the probability of transitioning from s_t to s_{t+1}.

$$\mathbb{P}(\tau|\theta) = p(s_o) \prod_t \mathbb{P}(s_{t+1}|s_t, a_t) * \pi(a_t|s_t) \tag{9}$$

Lets take the log of the above equation to use the log-derivative trick, which while helping us in converting the objective into an expectation, also simplifies it by converting this product into a summation which is easy to compute.

$$\log \mathbb{P}(\tau|\theta) = log(p(s_o)) + \sum_t \log \mathbb{P}(s_{t+1}|s_t, a_t) + \sum_t \log \pi(a_t|s_t) \tag{10}$$

Taking the gradient of this w.r.t θ so that it can be plugged into equation 8.

$$\Delta_\theta \log \mathbb{P}(\tau|\theta) = \Delta_\theta p(s_o) + \Delta_\theta \sum_t \log \mathbb{P}(s_{t+1}|s_t, a_t) + \Delta_\theta \sum_t \log \pi(a_t|s_t) \tag{11}$$

Since, the transition probability nor the state probability depend on theta(policy parameters) the gradient w.r.t theta will essentially be 0, giving us this simplified equation. The probability of a trajectory is replaced with the sum of probability of taking each actions in the trajectory which is nothing but the output of our policy π

$$\Delta_\theta \log \mathbb{P}(\tau|\theta) = \Delta_\theta \sum_t \log \pi(a_t|s_t) \tag{12}$$

Finally, lets plug the above equation into our expectation in equation 8.

$$\Delta_\theta J(\pi_\theta) = \mathbb{E}_{\tau \sim \pi_\theta}[\sum_t \Delta_\theta \log \pi(a_t|s_t) R(\tau)] \tag{13}$$

So, to solve Policy gradient using samples our final objective is the sum of gradient of the log of probability of each action multiplied by the final reward.

$$J(\pi_\theta) = \sum_t \Delta_\theta \log \pi(a_t|s_t) R(\tau) \tag{14}$$